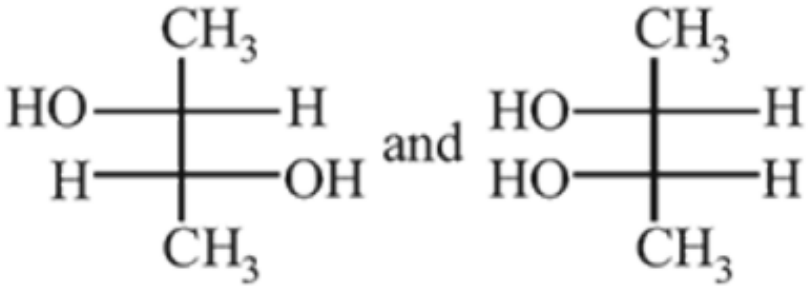
Quizrr Chapter-Wise Test for JEE Main - 2025

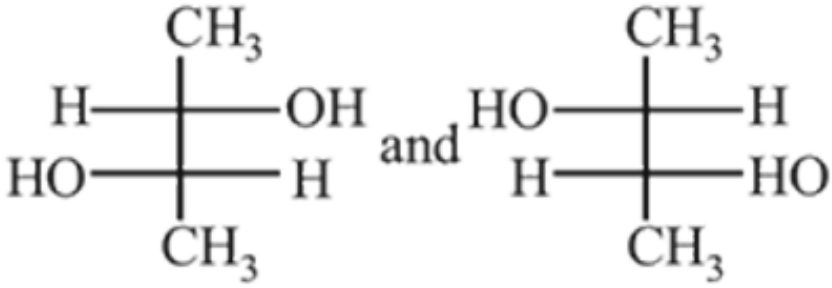
Q.1

Which of the following pairs of compounds are enantiomers?

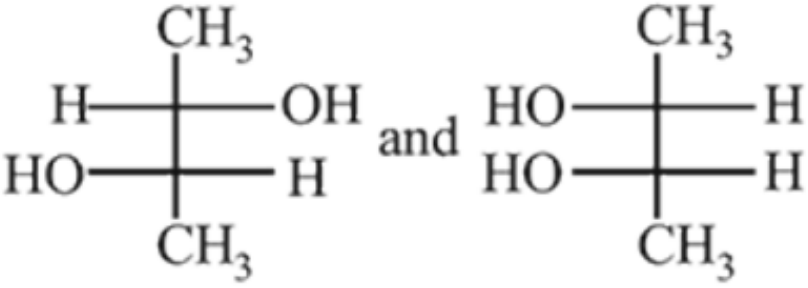
(A)



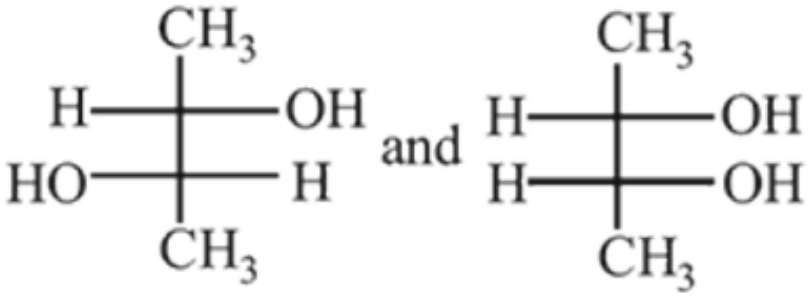
(B)



(C)



(D)



Q.2

Iodoform can be prepared from all except :

(A) Ethyl methyl ketone

(B) Isopropyl alcohol

(C) 3 -Methyl 2-butanone

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(D) Isobutyl alcohol

Q.3

In a nucleophilic substitution reaction:

$$R-Br+Cl^{-} \xrightarrow{DMF} R-Cl+Br^{-}$$

which one of the following undergoes complete inversion of configuration?

(A) $(C_6H_5)_2\,CH(Br)$

(B) $C_6H_5CH_2Br$

(C) $C_6H_5CH\,(CH_3)\,(Br)$

(D) $(C_6H_5)_2\,C\,(CH_3)\,(Br)$

Q.4

Benzene reacts with n -propyl chloride in the presence of anhydrous $AlCl_3$ to give

(A) $3-$ Propyl $-1-$ chlorobenzene

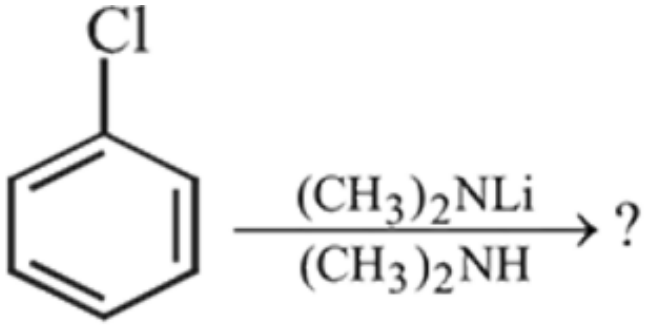
(B) n-Propylbenzene

(C) No reaction

(D) Isopropylbenzene

Q.5

What is the product of the following reaction ?



(A) N, N-dimethyl aniline

(B) Phenyl lithium (C_6H_5Li)

(C) para chloro-N, N-dimethyl aniline

(D) meta chloro-N, N-dimethyl aniline

Q.6

The major product formed when $1,1,1$ -trichloropropane is treated with aqueous potassium hydroxide is:

(A) Propyne

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(B) 1-Propanol

(C) 2-Propanol

(D) Propionic acid

Q.7

Consider the reactions:

(i) $(CH_3)_2\,CH - CH_2Br \xrightarrow{C_2H_5OH} (CH_3)_2\,CH - CH_2OC_2H_5 + HBr$

(ii) $(CH_3)_2\,CH - CH_2Br \xrightarrow{C_2H_5O^-} (CH_3)_2\,CH - CH_2OC_2H_5 + Br^-$

The mechanisms of reactions (i) and (ii) are respectively:

(A) S_N1 and S_N2

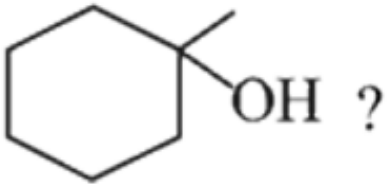
(B) S_N1 and S_N1

(C) S_N2 and S_N2

(D) S_N2 and S_N1

Q.8

How many stable alkenes on reaction with HCl will produce



?

(A) 2

(B) 3

(C) 1

(D) 4

Q.9

Number of ambidentate nucleophiles among the following

$CN^-, SCN^-, OH^-, CNO^-, SO_4^{2-}, HCO_2^-, NO_2^-, NO_3^-$

(A) 3

(B) 2

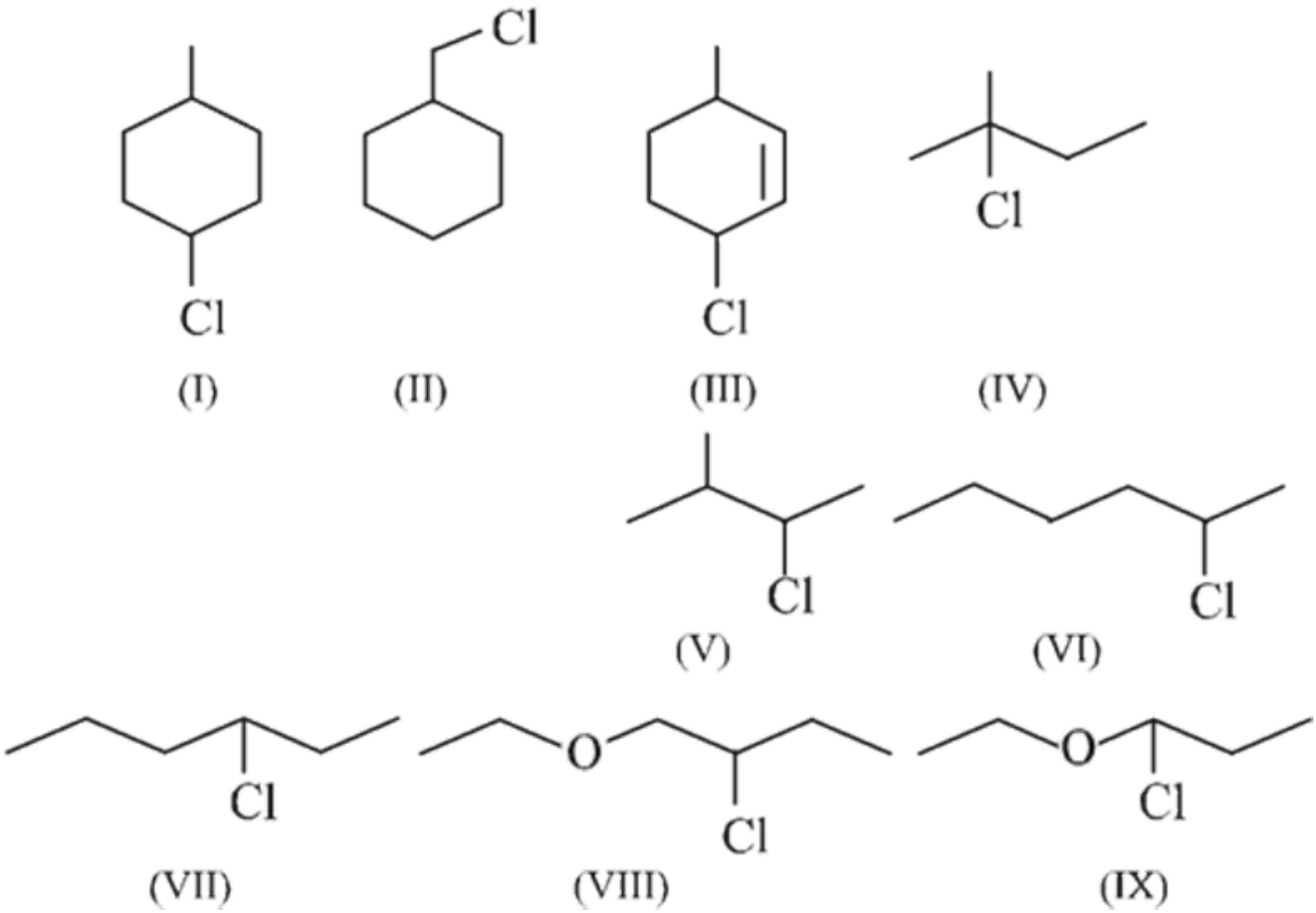
(C) 1

(D) 4

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Q.10

Number of substrates which will show rearrangement in



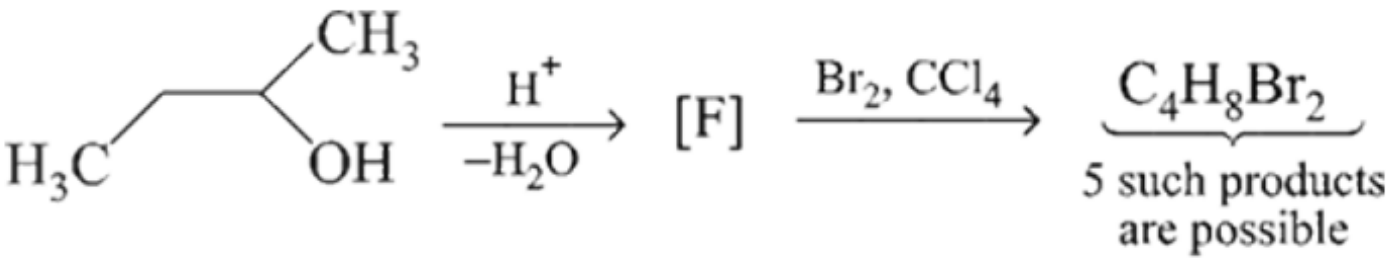
(A) 5

(B) 4

(C) 6

(D) 3

Q.11



How many structures for F are possible?

(A) 2

(B) 3

(C) 1

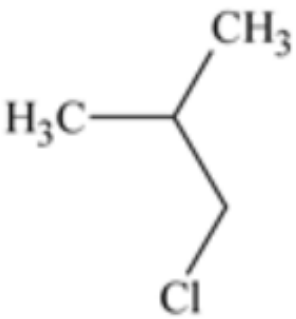
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(D) 4

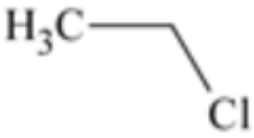
Q.12

Which one of the following compounds most readily undrgoes substitution by S_N2 mechanism?

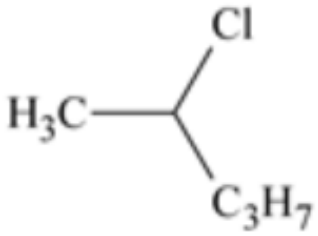
(A)



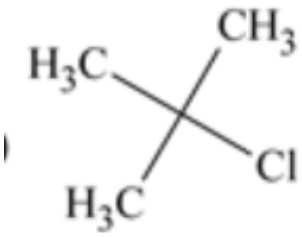
(B)



(C)



(D)



Q.13

Which of the following compounds would be hydrolysed most easily?

(A) C_2H_5Br

(B) CH_3Br

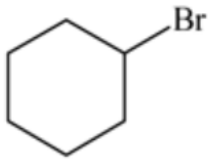
(C) $CH_2 = CH - Br$

(D) $CH_2 = CH - CH_2Br$

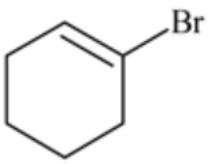
Q.14

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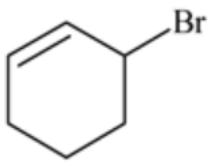
Which of the following is fast debrominated?



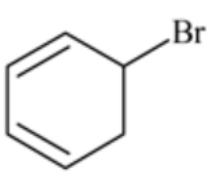
(I)



(II)



(III)



(IV)

(A) IV

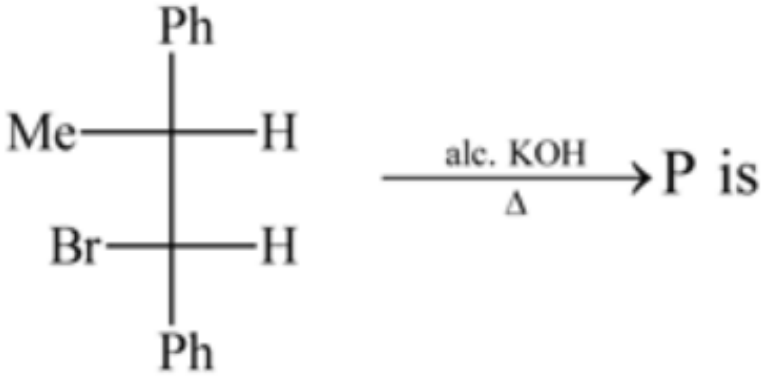
(B) II

(C) III

(D) I

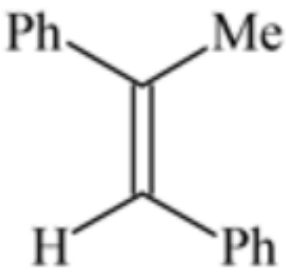
Q.15

The organic compound (P) of the reaction:

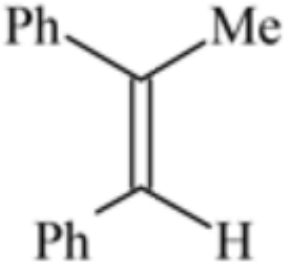


is

(A)

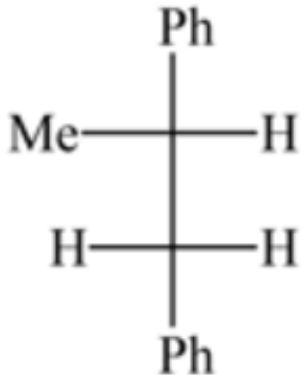


(B)

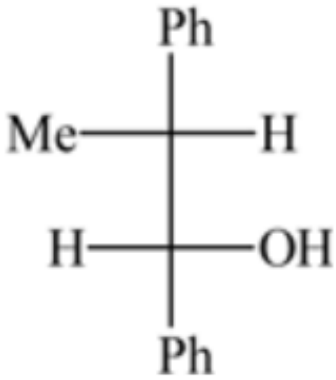


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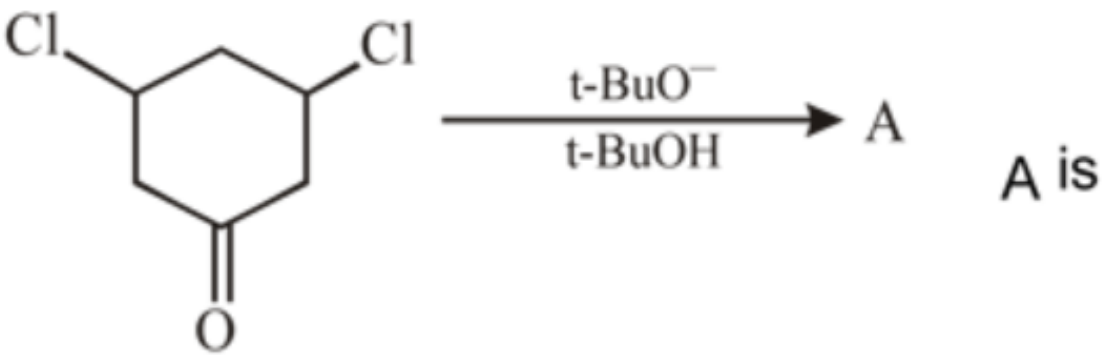
(C)



(D)

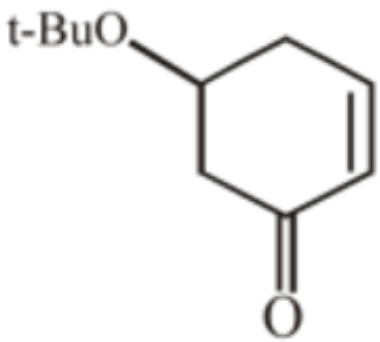


Q.16

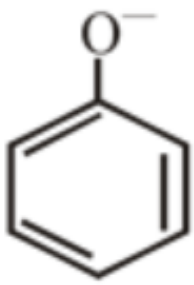


A is

(A)

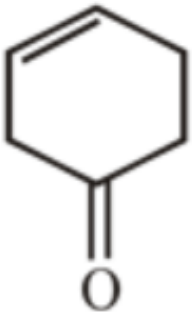


(B)

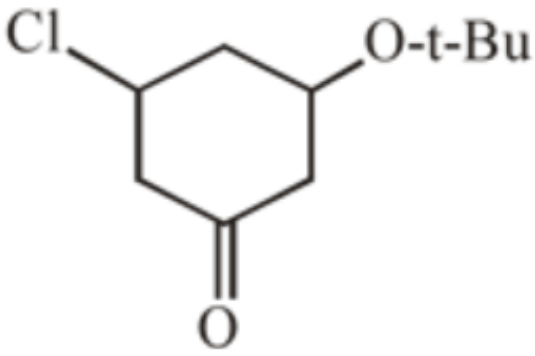


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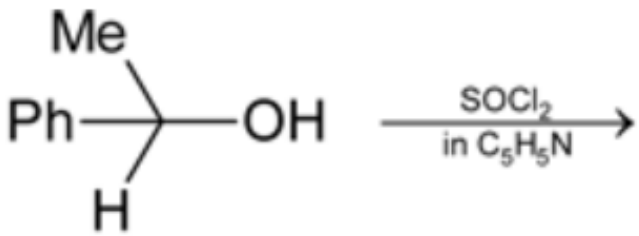
(C)



(D)



Q.17



Which statement is true for the above reaction?

(A) Retention of configuration

(B) Inversion of configuration

(C) Inversion and Retention both

(D) None

Q.18

On on heating diethyl ether with conc- HI, 2 moles of which of the following is formed

(A) Ethanol

(B) Iodoform

(C) Ethyl iodide

(D) Methyl iodide

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Q.19

Alkyl halide reacts with an alcoholic solution of ammonia to give a mixture of:

(A) 1° and 2° amine

(B) $1^\circ, 2^\circ, 3^\circ \&$ quarternary

(C) $1^\circ, 2^\circ \& 3^\circ$ amines

(D) $1^\circ \& 3^\circ$ amine

Q.20

Product-I $\xleftarrow{\text{aq.KOH}}$ C_2H_5Br $\xrightarrow{\text{alc.kOH}}$ Product-II The correct statement is

(A) product-I is obtained by the elimination reaction

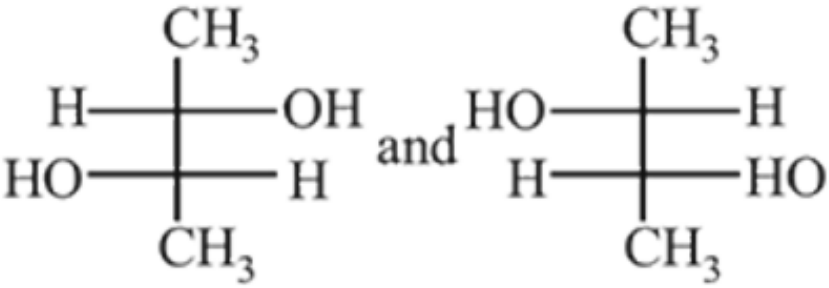
(B) product-II is obtained by the substitution reaction

(C) the molecular formula of product-I is $C_2H_{4'}$ while the molecular formula of product-II is C_2H_6O

(D) product-I is the isomer of dimethyl ether, while product-II is the dehydrated compound of product-I.

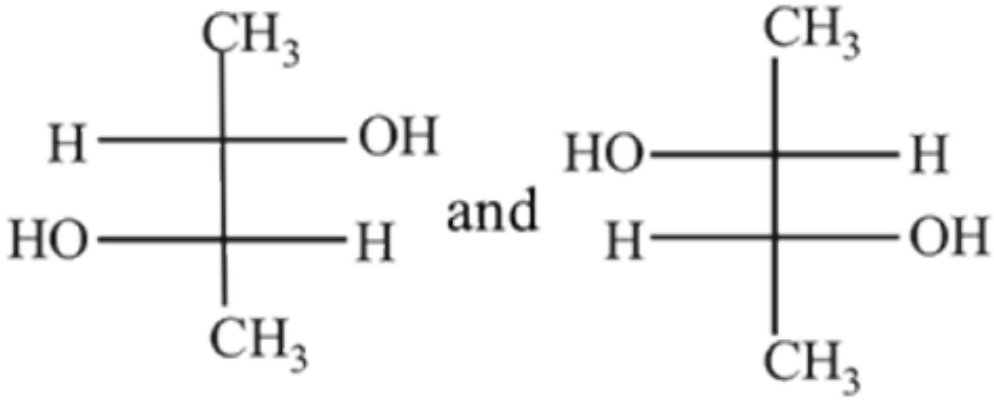
Answers & Solutions

Q.1 Answer:



Solution:

Compound which are mirror image of each other and are non superimposable are termed as enantiomers.



These are enantiomers

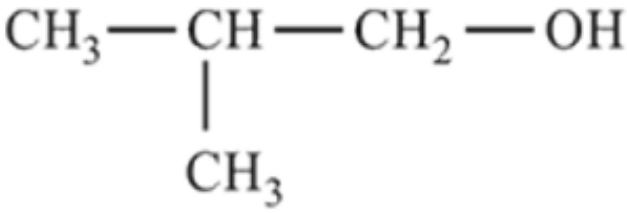
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Q.2 Answer:

Isobutyl alcohol

Solution:

Iodoform test is given by methyl ketones, acetaldehyde and methyl secondary alcohols.



Isobutyl alcohol is a primary alcohol hence does'nt give positive iodoform test.

Q.3 Answer:

$C_6H_5CH\,(CH_3)\,(Br)$

Solution:

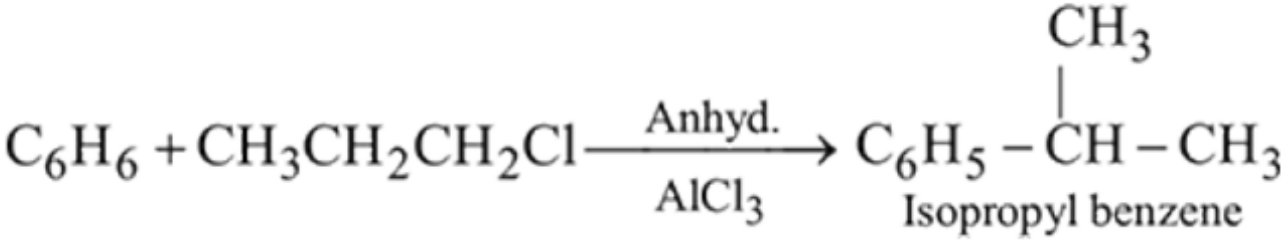
$C_6H_5CHCH_3Br$ **being an optically active secondary alkyl bromide undergoes** S_N2 **nucleophilic substitution reaction. Hence it undergoes complete inversion of configuration.**



Q.4 Answer:

Isopropylbenzene

Solution:

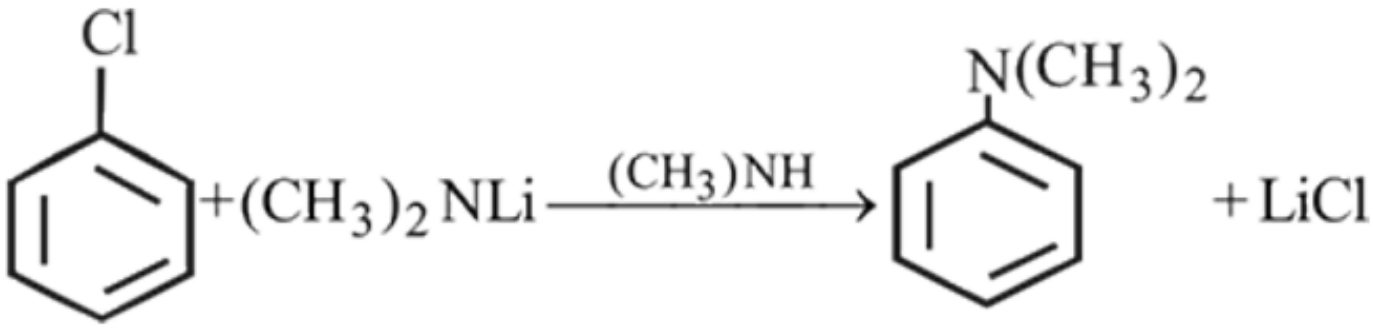


Q.5 Answer:

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N, N-dimethyl aniline

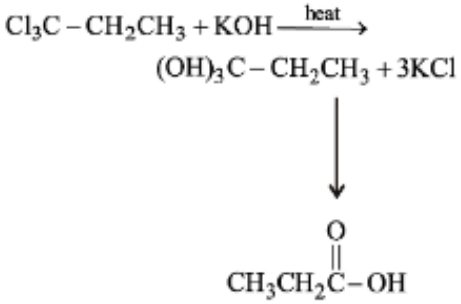
Solution:



Q.6 Answer:

Propionic acid

Solution:



$Cl_3CCH_2CH_3 + KOH \rightarrow (OH)_3CCH_2CH_3 + 3KCl$

— Now, as we can see that three molecules of hydroxide ion from potassium hydroxide replaces the chlorine atom and eliminates a by-product that is potassium chloride.

— But the presence of three hydroxyl groups causes the steric hindrance and also makes the molecule unstable.

— Due to the instability of the three hydroxyl groups, the product formed will release the water molecule due to which the formation of more stable carboxylic acid takes place.

-The balanced chemical will be as shown below:

$(OH)_3CCH_2CH_3 + KOH \rightarrow (OH)COCH_2CH_3 + 3H_2O$

— So, the final stable product which will be formed is propionic acid.

Q.7 Answer:

S_N1 and S_N2

Solution:

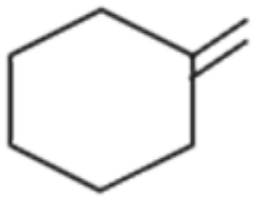
A strong nucleophile favours the S_N2 reaction and a weak nucleophile favours the S_N1 reaction. First reaction is S_N1 reaction because C_2H_5OH is used as solvent which is a weak nucleophile. Second reaction is S_N2 reaction because $C_2H_5O^-$ is strong nucleophile.

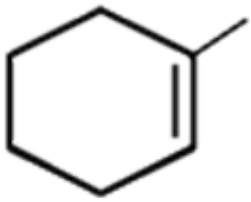
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Q.8 Answer:

2

Solution:





Most stable
(saytzeff's alkene)

Q.9 Answer:

4

Solution:

$CN^-, SCN^-, CNO^-, NO_2^-$

Q.10 Answer:

4

Solution:

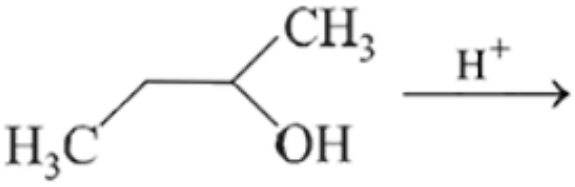
(II), (III), (V) and (IX) will show rearrangement.

Q.11 Answer:

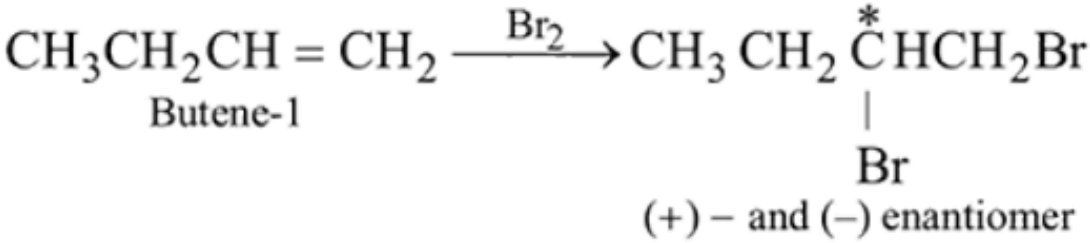
3

Solution:

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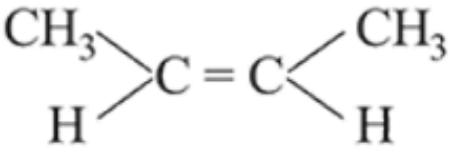


$$\underset{\text{Butene-1}}{CH_3CH_2CH=CH_2} + \underset{\substack{\text{Butene-2}\\(\textit{cis}\text{- and }\textit{trans})}}{CH_3CH=CHCH_3} \longrightarrow \underset{F}{\quad}$$

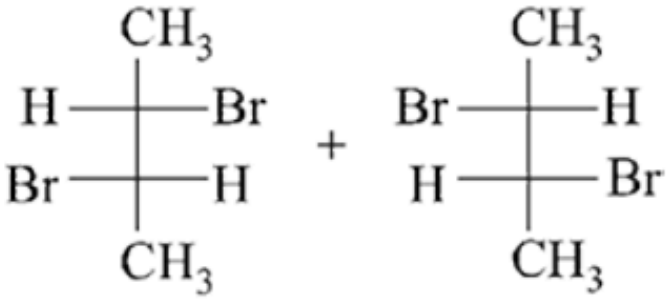


Butene-1

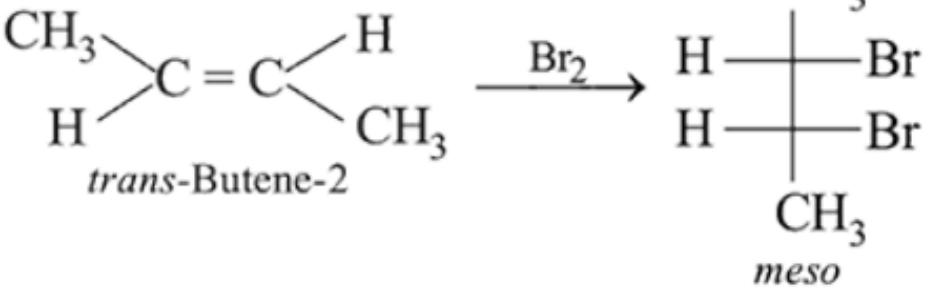
(+) – and (–) enantiomer



cis-Butene-2



Enantiomers



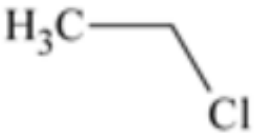
trans-Butene-2

meso

So F can have three possible structures

Q.12 Answer:

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Solution:

The increasing order of the nucleophilic substitution by S_N2 mechanism is tert $-$ alkyl halide $<$ sec alkyl halide $<$ primary alkyl halide.

Compound B is primary alkyl halide.

Hence it most readily undergoes substitution by S_N2 mechanism.

Note: compound A is also primary alkyl halide but in it, the carbon attached to is more hindered.

Q.13 Answer:

$CH_2 = CH - CH_2Br$

Solution:

In case of given compounde, Jdydralypio involves the breaking of C - Br bond. leading to the formation of Corbocation. $C^{(t)}$

If the formed carbocation is move stable, then rate of hydrolysis will be more.

$C_2H_5Br \longrightarrow C_2H_S^{\Theta}$

Methylcarbocation

$CH_3Br \longrightarrow CH_3\oplus$

1° **- carbocation**

$CH_2 = CH - Br \rightarrow CH_2 = CH^{\oplus}$

$CH_2 = CH - CH_2Br \rightarrow CH_2 = CH - CH_2^{\oplus}$

resonance stabilized carbocation.

The most stable carbocation is :

$CH_2 = CH - CH_2^{\oplus}$

Hence, $CH_2 = CH - CH_2$ br will be hydrolised most easily.

Q.14 Answer:

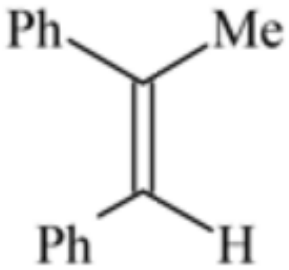
IV

Solution:

More the stabllity of tie product, faster is debromination of the parent compound.

Q.15 Answer:

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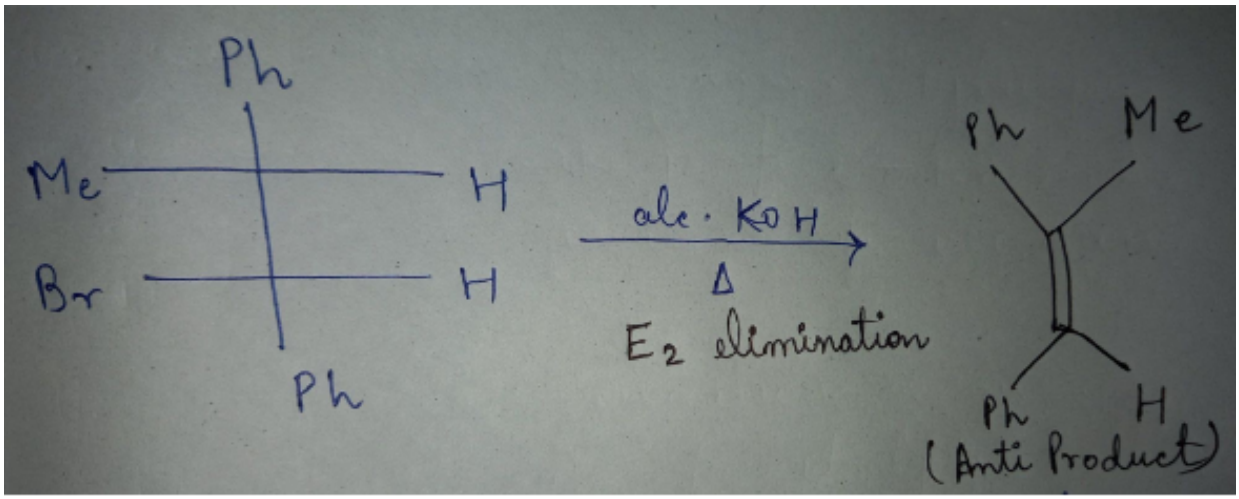
Solution:

We must know that the Haloalkane in the presence of alcoholic KOH undergoes elimination reactions. For example:

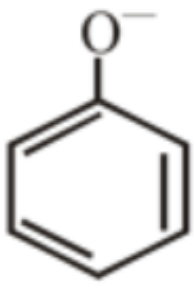
$$CH_3CH_2Br + KOH\ (alc.) \Rightarrow H_2C = CH_2 + KBr + H_2O$$

Alcoholic potassium hydroxide, KOH solution that functions as solvent gives alkoxide ions that act as a strong base. This base abstracts β-Hydrogen atom from saturated substrate - alkyl halide. Abstracted β-Hydrogen atom is then transferred to the alkyl part to form an alkane and simultaneously a molecule of HCl is eliminated.

The basicity of hydroxide ions is considerably lower than the basicity of alkoxide ions. Hydroxide ion is significantly getting hydrated in aqueous solution. Hence, hydroxide ions cannot abstract β-hydrogen atoms of alkyl chloride to eliminate HCl and form alkene.

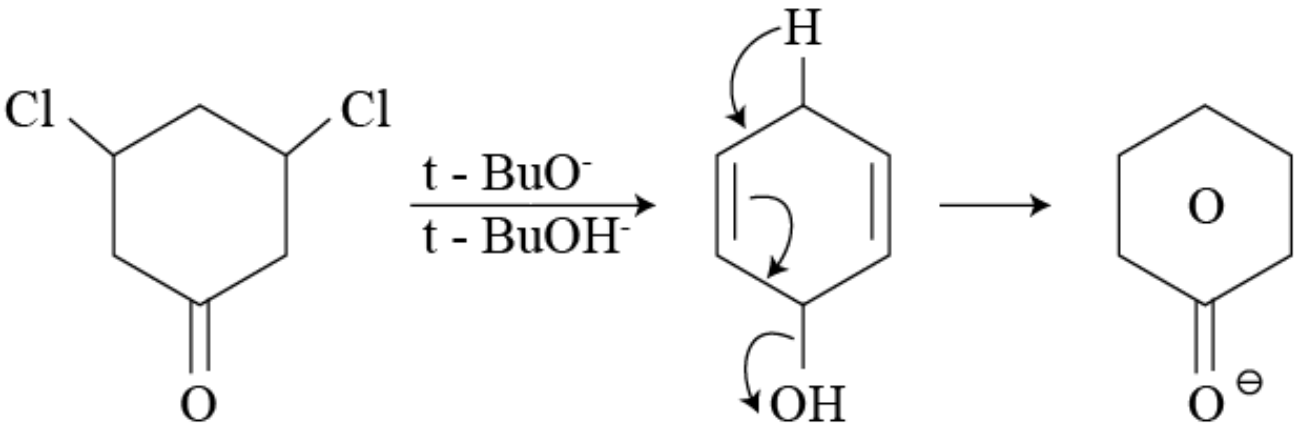


Q.16 Answer:



Solution:

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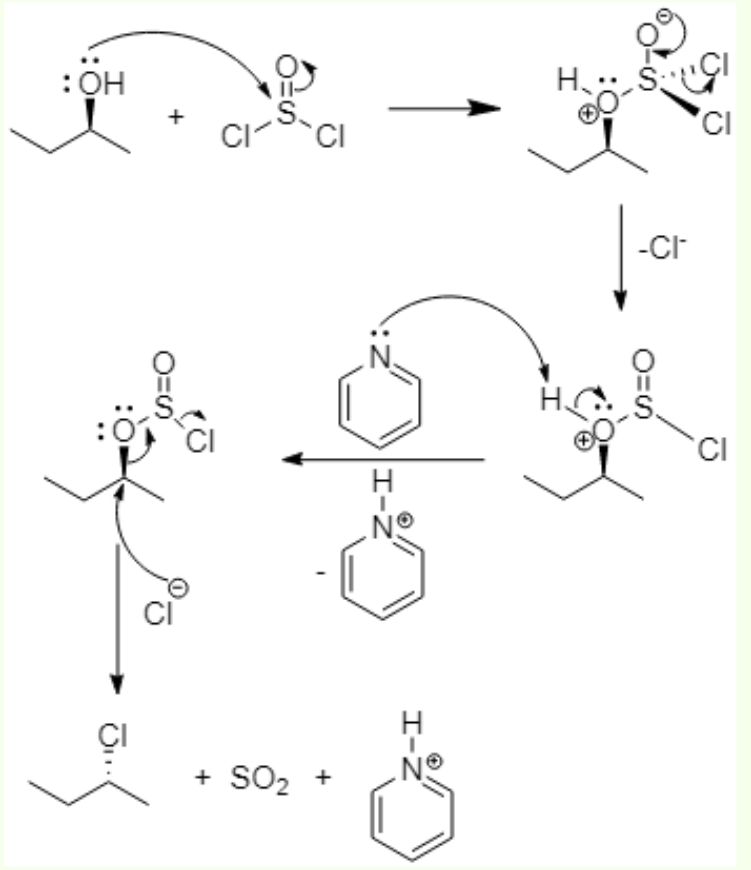


Q.17 Answer:

Inversion of configuration

Solution:

The correct option is B Inversion of configuration.



Here inversion of configuration occurs.

Q.18 Answer:

Ethyl iodide

Solution:

Firstly the HI given will get ionized:

$$HI \rightarrow H^+ + I^-$$

The iodide ion produced will attack one of the Carbon-oxygen bonds, gets attached to carbon, the hydrogen ion produced in the first step will get attached to a negatively charged oxygen atom:

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$CH_3CH_2O - CH_2CH_3 + HI \rightarrow CH_3CH_2OH + I - CH_2 - CH_3$

The ethanol produced in the above step still has a carbon-oxygen bond and we are provided with an excess of HI (concentrated), so the iodide ion will now attack the carbon-oxygen bond of ethanol similarly, and the hydrogen ion will get attached to hydroxide ion produced:

$CH_3CH_2 - OH + HI \rightarrow CH_3CH_2 - I + H_2O$

Thus, the overall reaction between diethyl ether and hot and concentrated HI will be:

$CH_3CH_2OCH_2CH_3 + 2Hl \rightarrow 2CH_3CH_2I + H_2O$

Means, the given reaction will give two moles of ethyl iodide.

Q.19 Answer:

$1^{\circ}, 2^{\circ}, 3^{\circ} \&$ quarternary

Solution:

Acoholic solution of ammonia is heated in a sealed tube at $100^{\circ}C$

$$RI \xrightarrow{NH_3} \underset{1^{\circ}\text{ amine}}{RNH_2} \xrightarrow{RI} \underset{2^{\circ}\text{ amine}}{(R)_2NH} \xrightarrow{RI} \underset{3^{\circ}\text{amine}}{(R)_3\,N} \xrightarrow{RI} \underset{4^{\circ}\text{amine}}{(R)_4N^+I^-}$$

Q.20 Answer:

product-I is the isomer of dimethyl ether, while product-II is the dehydrated compound of product-I.

Solution:

$C_2H_5Br \xrightarrow{aqKOH} C_2H_5OH$ **(Product-I)**

(Nucleophilic substitution reaction)

$C_2H_5Br \xrightarrow{deKOH} C_2H_4$ **(Product-II)**

(elimination reaction)

C_2H_5OH **(Product-I) isomer is** CH_3OCH_3

C_2H_5OH **denydration** $\longrightarrow C_2H_4$

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